

CSE 776: Deep Learning for Computer Vision (January 2026)

(First Course Handout)

Course Objective: This course will be focused on introducing students to computer vision and foundations of deep learning (DL) models to improve computer vision (CV). It will start with image preprocessing, basic architecture of convolutional neural networks and will progress to advanced CNN models for image classification and object detection. The course will heavily emphasize on a course project to give students hands-on coding experience of developing CV applications.

Instructor: Priyanka Bagade

Class venue: RM101 (Tuesday 12:00 noon to 01:15pm and Wednesday 12:00 noon to 01:15pm)

Prerequisites: Strong understanding of basic probability, calculus and linear algebra is required. The students will be expected to have good programming skills.

Course Contents: Here is a broad list of topics to be covered in the course.

S. No	Broad Title	Topics
1	Introduction to Computer Vision	Applications, CV pipeline, image preprocessing, feature extraction
2	Deep Learning models	Concept of perceptron, activation functions, feedforward process, backpropagation, optimization algorithms
3	Convolutional Neural Networks (CNN)	CNN architecture, basic components of CNN< Image classification with CNN, optimizations
4	Model performance evaluation	Performance metrics, model evaluation, improving model, hyperparameter tuning, optimizing algorithms
5	Advanced models in CNN for classification and detection	LeNet-5, AlexNet, VGGNet, ResNet, R-CNN, SSD, YOLO
6	Transfer Learning	Basics of transfer learning, approaches, open source datasets
7	Generative adversarial networks (GAN)	GAN architecture, model evaluation

- The course will also involve reading related research papers

Book/References:

1. Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville
2. Deep learning for Vision Systems by Mohamed Elgendy, Manning publisher
3. Research papers from top-tier conferences and journals.

Grading Scheme:

- Attendance (5%)
- Quiz (20%)
- Assignment (10%)
- Mid semester exam (25%)
- Course project (40%)

This is a tentative allocation.

- Might change depending on the strength of the class and class progress
- Grading will be relative